Estela Pérez Luque, PhD

Human Behaviour & Model Validation Engineer

+46 73 262 8876 • perezluque.estela1504@gmail.com • LinkedIn • Portfolio • Skövde, Sweden

PROFILE

I work at the intersection of human behaviour, data, and model validation. My background combines rigorous evaluation of predictive models with real-world data collection, turning complex time-series data into reliable, decision-ready evidence. I am particularly driven by understanding how people behave, including variability, diversity, and context, and translating that into technology that is robust, trustworthy, and useful in practice. My goal is to bridge the gap between model performance and real-world impact, ensuring that systems interpreting human behaviour are meaningful and reliable across real-world environments.

TECHNICAL SKILLS

Core tools: MATLAB, Python, Excel

Methods: V&V, simulation-based evaluation, model benchmarking, data analysis, time-series data, motion capture

Domain: Human behaviour, automotive/transportation systems, experimental data

Additional tools: IPS IMMA, Siemens Jack, Autodesk Inventor, SolidWorks

Languages: English (C1), Spanish (Native), Swedish (B1-improving)

PROFESSIONAL EXPERIENCE

PhD Researcher - University of Skövde, Sweden (Onsite) **Sep 2020 – Oct 2025**

- Built end-to-end MATLAB and Python pipelines to benchmark predictive models against motion capture and in-vehicle datasets, analysing datasets of up to 199 participants across multiple tasks and operational conditions, and generating reproducible, traceable evidence to support model validation and design decisions in automotive contexts.
- Designed and executed scenario-based validation studies comparing simulated and real-world behaviour across multiple vehicle configurations and user groups; enabled structured model comparison and reduced reliance on subjective assessment in early design stages.
- Developed quantitative evaluation frameworks to assess agreement between predicted and observed behaviour, identifying up to ~15–20% variation in model performance depending on evaluation approach, and demonstrating how evaluation choices directly impact conclusions in applied contexts.
- Applied multi-objective optimization (NSGA-II) to evaluate 1,000+ vehicle interior configurations across 14 digital human models, transforming ergonomics assessment from manual trial-and-error into a scalable, data-driven design exploration process.
- Collaborated with Volvo Cars, Volvo Trucks, Scania, and the SAFER vehicle safety ecosystem, aligning validation strategies with engineering constraints and real-world use cases; enabled faster, more consistent decision-making based on data rather than assumptions.
- Delivered research outcomes through 20+ scientific publications and 15+ presentations (international conferences, industry workshops, and invited talks), translating complex model evaluation into decision-ready insights and increasing adoption of data-driven validation approaches across research and industry stakeholders.

Visiting Researcher - University of Antwerp, Belgium (Onsite) **Sep 2024 – Mar 2025**

- Led full-cycle experimental data collection for a large-scale, multi-year study, including ethics approval, protocol design, participant recruitment, and multi-sensor data acquisition, enabling long-term, structured validation of human behaviour models across datasets and studies.

Estela Pérez Luque, PhD

Human Behaviour & Model Validation Engineer

+46 73 262 8876 • perezluque.estela1504@gmail.com • LinkedIn • Portfolio • Skövde, Sweden

- Designed and executed motion capture and 4D body scanning protocols for 30+ participants, ensuring consistent data quality and reproducibility across experimental conditions, improving reliability of validation datasets used for model evaluation.
- Built structured, reproducible data pipelines to standardize multi-site data collection and processing; increased data consistency and comparability across institutions, supporting more robust cross-study validation.

Co-founder and R&D Lead - Swe-Ex AB, Sweden (Startup) **Sep 2021 – Mar 2023**

- Co-founded a startup developing wearable safety support equipment for industrial environments, focusing on ergonomic risk reduction and worker support.
- Translated user needs and safety requirements into measurable design and validation criteria, guiding three full prototype iterations from CAD concept to physical validation using 3D-printed, textile, and steel components.
- Developed and tested prototypes using structured physical validation methods, enabling rapid iteration cycles and evidence-based design decisions.
- Built business and product understanding through a 2-year incubator program, gaining experience in stakeholder communication, value proposition development, and prioritization under uncertainty.

Research Assistant - University of Skövde, Sweden (Onsite) **Jun 2019 – Sep 2020**

- Developed and executed simulation studies and ergonomic analyses across multiple projects with 6+ industry partners, producing structured reports and live demonstrations, and enabling stakeholders to understand and act on simulation results in early design phases.
- Created educational materials and training content, including guided video tutorials, for industrial partners and students, supporting adoption of digital human modelling tools and improving usability and onboarding time, and increasing effective use of simulation tools in both academic and industry settings.

EDUCATION

PhD in Informatics, University of Skövde (Sweden) Sep 2020 – Oct 2025

PhD Thesis: Human Posture and Motion Prediction for Automotive Ergonomics Design

Focus: Human behaviour, model validation, time-series analysis, simulation benchmarking

Master of Science in Intelligent Automation, University of Skövde (Sweden) Aug 2018 – Jun 2019

Bachelor of Science in Mechanical Engineering, University of Skövde (Sweden) Aug 2017 – Jun 2018

Degree in Mechanical Engineering, University of Málaga (Spain) Aug 2012 – Jun 2018

RESEARCH PUBLICATIONS

Full publication list available at: Google Scholar – Estela Pérez Luque